

Principles of sterilization

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KEY POINTS

- Many dosage forms, medicinal products and devices need to be free of living microorganisms.
- Failure to achieve sterility and the lack of validation or documentation of a sterilization process have led to patient deaths.
- Pharmacopoeias usually recommend five processes for the sterilization of sterile dosage forms: steam (under pressure), dry heat, gaseous (ethylene oxide), ionizing radiation and filtration sterilization.
- The use of sterilization parameters allows the calculation of the efficacy of a given sterilization regimen for a

given load/product and enables comparison of efficacy between processes.

- High-level disinfection is used for the 'sterilization' of certain medical devices.
- A number of new technologies, notably high-pressure and gas plasma sterilization, might offer appropriate alternatives to the established sterilization processes.

Introduction

Previous chapters have described the types and properties of microorganisms (see [Chapter 13](#)) and the action of heat and chemical agents on them (see [Chapter 15](#)). This chapter will build on those fundamentals and describe the principles underlying the different methodologies available to achieve sterility. These will be described both for pharmaceutical preparations and for medicinal products and devices. This chapter will also describe the criteria used to measure sterility. The practicalities associated with the processes of sterilization are described in [Chapter 17](#).

By definition, a sterile preparation is described as the absolute absence of viable microbial contaminants. In practice, this definition is not achievable as a preparation cannot be *guaranteed* to be sterile. This remark is discussed further in [Chapter 17](#).

Certain pharmaceutical preparations, medical devices and items for which their use involves contact with broken skin, mucosal surfaces or internal organs, injection into the bloodstream and other sterile parts of the body are required to be sterile. These are frequently referred to in pharmacopoeias as sterile products or sterile dosage forms. Microbiological materials, such as soiled dressings and other contaminated items, also need to be sterilized before disposal or reuse.

Sterilization is the process by which a product is rendered sterile, i.e. by the destruction or removal of

microorganisms. The majority of the processes recommended by pharmacopoeias (i.e. steam under pressure sterilization, dry heat sterilization, gaseous sterilization and ionizing radiation sterilization) are terminal sterilization processes, whereby the preparation is sterilized in its final container or packaging. For other multiple-component preparations that cannot be sterilized with such methods, filtration sterilization can be used. Finally, high-level disinfection is used for the 'sterilization' of medical devices.

Need for sterility

As indicated in the introduction, certain pharmaceutical preparations, medicinal products and devices are required to be sterile (further information is given in [Table 17.1](#)). Briefly, these include:

- injections – intravenous infusions, parenteral nutrition (PN) fluids, small-volume injections and small-volume oily injections;
- noninjectable sterile fluids – noninjectable water, urological irrigation solutions, peritoneal dialysis and haemodialysis solutions, nebulizer solutions;
- ophthalmic preparations – eye drops, lotions and ointments and some contact lens solutions;
- dressings;
- implants;
- absorbable haemostats;
- surgical ligatures and sutures (absorbable and nonabsorbable); and
- instruments and equipment – syringes, metal instruments, respirator parts, medical devices, endoscopes.

Failure to achieve sterility can result in serious consequences. In the best-case scenario, surviving microorganisms induce spoilage of the product (i.e. chemical and physical degradation) that might be identified before the preparation is used. The product (or batch of product) is then removed from use and destroyed. In the worst-case scenario, where microbial survival cannot be identified through deleterious effects on the product, infection of patients (sometimes fatal) might result from the use of the contaminated preparation. There have been many reports in the literature of such incidents over the years. For example, in the Devonport Incident (in 1971–1972), the death of five patients (from acute endotoxic shock) was traced to dextrose 5% infusion bottles 'sterilized' using a faulty autoclave. Other examples include: In 1996, in Romaira (Brazil), 35 newborn infants died of sepsis attributed to locally produced intravenous solutions; in 2005, a contaminated heparin intravenous flush was responsible for infecting several patients in different states

in the USA and in 2012, methylprednisolone acetate epidural injections contaminated with *Aspergillus fumigatus* were responsible for multiple fatalities. In these cases, inappropriate manufacturing and quality assurance procedures were implicated. These incidents emphasize that not only must an appropriate sterilization regimen be used but appropriate monitoring and control must also be performed. This requires an understanding of the general principles of Good Manufacturing Practice (GMP; see [Chapter 54](#)) and in particular the principles of sterilizing processes and their control and validation.

Sterilization parameters

The inactivation kinetics of a pure culture of microorganisms exposed to a physical or chemical sterilization process is generally described by an exponential relationship between the number of organisms surviving and the extent of treatment ([International Organization for Standardization, 2009](#)), although variations from this are likely ([Chapter 15](#) gives more details). Survivor curves have been used to generate inactivation data for specific sterilization processes using specific biological indicators (see [Chapter 17](#)). These data are important for the calculation of a number of sterilization parameters which help to establish a sterilizing regimen adapted to a specific preparation or product.

D value and Z value

One of the important concepts in sterilization is the *D* value ([Fig. 16.1](#)). This parameter is calculated as the time taken to achieve a 1-log (90%) reduction in the number of microorganisms. Another important concept is the *Z* value, which represents the increase in temperature for steam (under pressure) or dry heat sterilization, or the dose for radiation sterilization, required to produce a 1-log (90%) reduction in the *D* value for a particular microorganism. This parameter is used to compare the heat (or dose) resistance of different biological indicators following alterations in temperature or radiation exposure. [Chapter 15](#) provides more information on both these parameters.

Inactivation factor and most probable effective dose

The inactivation factor is the total reduction in the number of viable microorganisms brought about by a defined sterilization process. This parameter can be calculated from the *D* value but only if the destruction curve follows the linear logarithmic model. To overcome problems caused by variations from this model, a most probable effective dose

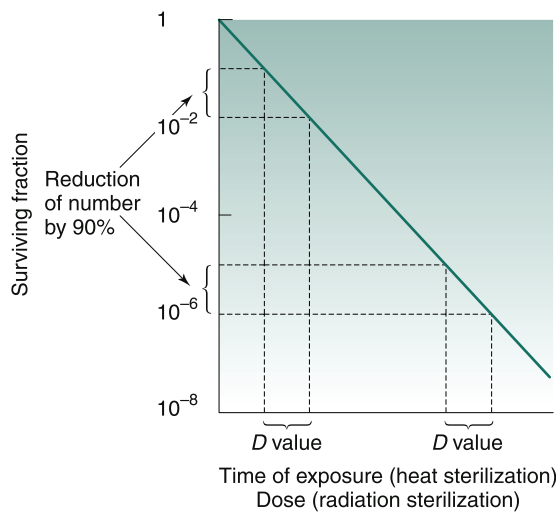


Fig. 16.1 Calculation of the D value. Note that the D value remains the same although it is calculated with different surviving fractions.

can be used. This is the dose needed to achieve n decimal reductions in the number of microorganisms.

F value

The F value is a measure of the total lethality of a heat sterilization process for a given microorganism and is used to compare the lethality of different heat sterilization processes. A reference value (F_0) of *Geobacillus stearothermophilus* spores at 121 °C is often used with a Z value of 10 °C. The total F_0 of a process includes the heating up and cooling down phases of the sterilization cycle.

For dry heat sterilization, the F value concept has some limited application. The F_H value is used and corresponds to the lethality of a dry heat process in terms of the equivalent number of minutes of exposure at 170 °C. A Z value of 20 °C is used for the calculation.

Principles of sterilization processes

Five main types of sterilization processes are usually recommended for pharmaceutical products ([British Pharmacopoeia Commission, 2020](#)). Among these, steam sterilization (sometimes referred to as steam under pressure sterilization or high-temperature steam sterilization) still represents the gold standard. Novel sterilization processes are being developed and have already been applied in the food industry.

These are mentioned in the section entitled 'New technologies' later in this chapter.

Heat sterilization

Heat has been employed as a purifying agent since early historical times, and is now used worldwide in sterilization. Boiling is not a form of sterilization as higher temperatures are needed to ensure the destruction of all microorganisms.

Microorganisms vary in their response to heat. Species of bacterial spores are thought to be some of the most heat-resistant forms of life and can survive temperatures above 100 °C. Nonsporulating bacteria are destroyed at lower temperatures (50 °C to 60 °C) and vegetative forms of yeasts and moulds have a similar response. Cysts of amoebae (e.g. *Acanthamoeba polyphaga*) are less sensitive than their vegetative cells, which are inactivated at 55 °C to 60 °C. It is generally thought that viruses are less resistant than bacterial spores. The agents responsible for spongiform encephalopathies, the prions, are worth mentioning due to their infectious nature and high resistance to heat (current thermal sterilization procedures are not effective in inactivating prions; see [Chapter 15](#)).

Despite the widespread use of heat sterilization, the exact mechanisms and target sites within microorganisms are still uncertain. It is likely that several mechanisms and targets are implicated and those proposed include damage to the outer membrane (Gram-negative bacteria) and cytoplasmic membrane, RNA breakdown and coagulation, damage to DNA and denaturation of proteins, probably as a result of an oxidation process. For hydrated cells (steam sterilization), it is likely that the chemically lethal reactions occur more rapidly in the presence of water. The denaturation and coagulation of key enzymes and structural proteins probably result from a hydrolytic reaction.

The thermal death of bacterial cells and spores is usually thought to have first-order reaction kinetics (see [Chapter 7](#)). Although some controversy over this exists, the use of an exponential inactivation model for the kinetics of spores is unlikely to underestimate the heat required. One way to express the rate of death as a first-order reaction is

$$N_t = N_0 e^{-kt} \quad (16.1)$$

This relationship is discussed further in [Chapter 15](#).

Heat sterilization processes usually occur in three phases:

- the heating up phase, where the temperature within the chamber of the sterilizer is brought to the appropriate level;
- the holding phase – when the optimal temperature is reached, the holding time is maintained for the required duration (e.g. 15 minutes at 121 °C); and

- the cooling down phase, where the chamber temperature is brought down before the preparation/product can be removed safely from the sterilizer.

Principles of steam sterilization

Steam sterilization relies on a combination of steam, temperature and pressure. Steam is used to deliver heat to the product to be sterilized. There are different types of steam, but only steam at the phase boundary (between water and steam) has the appropriate characteristics for maximum effectiveness (Fig. 16.2). Steam at the phase boundary between itself and its condensate has the same temperature as the boiling water that produced it but holds much more latent heat. This latent heat is available for transfer (without a decrease in temperature) when it condenses onto a cooler surface. This rapid transfer of latent heat is responsible for the rapid rise in temperature (to the sterilization temperature) of any items it touches (Chapter 15 provides more information). The condensed water also aids the process by hydrating microorganisms and making them more sensitive. The condensation of steam contracts it to a very small volume that creates a pressure decrease into which more steam is drawn to reestablish the pressure. This aids the penetration of steam into porous items such as dressings. Wet saturated steam is less effective than dry saturated

steam as not as much condensation is produced and the latent heat available is less (see Fig. 16.2). It can also saturate loads with free water and interfere with steam penetration. Superheated steam is another potential problem that must be limited (see Chapter 15). Although it is hotter than dry saturated steam, superheated steam is less efficient at releasing its heat to cooler objects, as it is only as efficient as hot air at the same temperature.

Principles of dry heat sterilization

Sterilization using dry heat is less efficient than sterilization using moist heat but is the preferred method for items that are thermostable but moisture sensitive or impermeable to steam. This includes some metal devices, glassware, oils/oily injections and some powders (see Table 17.1). In addition, temperatures greater than 220 °C are used for depyrogenation of glassware (in this case the process needs to demonstrate a 3-log reduction in the amount of heat-resistant endotoxin). The main advantage of dry heat sterilization over steam sterilization is its ability to penetrate items and kill microorganisms via oxidation. The process is generally slower and the holding times for dry heat sterilization are much longer than those needed for steam sterilization (Table 16.1).

Combination treatments

The amount of heat required for sterilization can be reduced by use of a combination treatment of heat plus a reduced pH (<4.5) or low water potential (A_w). Although some spores

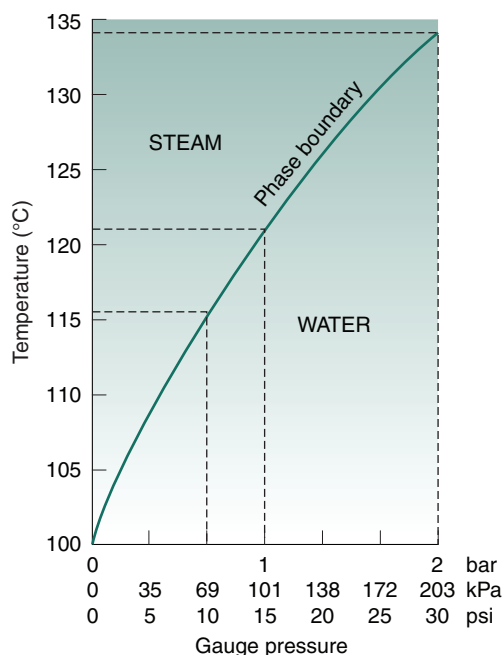


Fig. 16.2 Saturated steam phase diagram for moist heat sterilization.

Table 16.1 Time and temperature combinations used for moist and dry heat sterilization

Process	Minimum temperature (°C)	Minimum holding period (min)
Steam sterilization (autoclaving)	115	30
	121	15
	126	10
	134	3
Dry heat	140	180
	150	150
	160	120
	170	60
	180	30



Fig. 16.3 Endoscope washer from Steris Corporation.

may survive the initial treatment, they then germinate and die, reducing the numbers of organisms present as time passes (a process called autosterilization). This technique has been used in the food industry with meat-based canned foods to avoid high-temperature sterilization. In addition, the combination of heat and hydrostatic pressure is thought to be synergistic. This is further discussed in the section entitled 'Ultrahigh pressure' later in this chapter.

In the pharmaceutical industry, another type of combination treatment involves low-temperature steam formaldehyde (LTSF). The steam is below atmospheric pressure (70 °C to 80 °C) and formaldehyde gas provides the sporicidal effect. Formaldehyde has been used to sterilize medical devices but care must be taken with its use as it is acutely toxic to humans and is also mutagenic and possibly carcinogenic. Some caution should be attached to the sterility assurance level (see [Chapter 17](#)) of the process, as it has been suggested that revival of spores after LTSF treatment is possible ([McDonnell, 2007](#)). Finally, the processing of thermolabile medical equipment such as endoscopes is conducted in automated washer disinfectors ([Fig. 16.3](#)). These make use of high-level disinfectants and slightly raised temperature. This is further discussed in the section entitled 'High-level disinfection' later in this chapter.

Alternative means for heat delivery and control

Systems for the delivery of heat via direct application of flame, heating using alternating electric currents and

microwave energy have been developed and adopted in some parts of the world. The use of infrared radiation allows a rapid elevation of temperature. Advances in packaging have improved heat penetration and made it more uniform. These include flexible pouches, polypropylene rigid containers, thermoformed containers, tin-free steel and foil-plastic combinations.

Gaseous sterilization

When sterilization by heat is not possible, one alternative is to use a sterilizing gas. Not many gases are used in the pharmaceutical industry for sterilization, and it is important to note that some of these gases are also used for disinfection (as either a liquid or a gas) under different conditions. Pharmacopoeias usually recommend the use of ethylene oxide for gaseous sterilization of pharmaceutical preparations, although other chemicals are available, such as formaldehyde, hydrogen peroxide, chlorine dioxide, peracetic acid and ozone. The chemical biocides are generally divided, according to their mode of action, into alkylating and oxidizing agents.

Alkylating gases

Some alkylating gases can permeate through many polymeric materials and are therefore not limited to just surface applications.

Ethylene oxide is widely used in pharmaceutical manufacturing but less so in hospitals. It occurs in gaseous form at room temperature (boiling point 10.7 °C) and penetrates narrow spaces well. Ethylene oxide has been shown to possess bactericidal, fungicidal, virucidal, sporicidal and protozoacidal properties.

The use of LTSF sterilization has been mentioned already. Formaldehyde is a surface sterilant only and cannot be used to sterilize occluded areas. Penetration into porous materials can be inhibited by the formation of polymers that cross-link, preventing further sterilant access ([Chapter 15](#) provides more information).

Oxidizing gases

Oxidizing gases are relatively unstable, and their decomposition can lead to microenvironments within the load that are not exposed to the full concentration of the agent.

Hydrogen peroxide gas has been shown to be effective against spores at a range of temperatures. Its action is greatest when used at near-saturation levels on clean dry surfaces, and it does not leave a toxic residue. The vapour is usually obtained via evaporation of a heated stock solution. Water condensation on surfaces that are being sterilized can reduce local concentrations of hydrogen peroxide and hence its activity; decomposition by catalytic activity and

absorption by cellulosic materials (e.g. paper) can also reduce its effectiveness. The combination of hydrogen peroxide with cold plasma, cupric or ferric ions, ozone or ultraviolet radiation has been shown to enhance activity.

Ozone has been demonstrated to be an effective sterilant, but the complex control of humidity required and corrosion problems have limited its applications, and it is not routinely used. Ozone as a disinfectant for water currently shows more promise, and commercial systems are available for several applications.

Chlorine dioxide is a broad-spectrum biocide with a sporicidal activity and is mainly used for the high-level disinfection of medical devices. Its applications are limited due to its effect on materials such as uncoated aluminium foil, uncoated copper, polycarbonate and polyurethane.

Peracetic acid exists as a liquid at room temperature and is used for high-level disinfection (discussed later). Vaporized peracetic acid can be used for the sterilization of surfaces and devices, although a long contact time is required. Peracetic acid can cause corrosion of certain metals and rubbers and has low penetrating power.

Vaporized oxidizing agents are often used in combination or with plasma. Formulations also contain excipients that reduce their negative impact, such as smell and corrosiveness.

Radiation sterilization

There are two main types of radiation: electromagnetic and particulate.

- electromagnetic radiation — γ -rays, X-rays, ultraviolet radiation (UV), infrared radiation (IR), microwave energy and visible light
- particulate radiation — α -particles, β -particles (high-speed electrons), neutrons and protons.

Of these, only γ -rays and high-speed electron beams are used for sterilization of pharmaceutical products, since other forms of radiation have not been shown to be effective as sterilants and/or are not suitable. Both γ -rays and β -particles are forms of ionizing radiation (the properties of each type of radiation are given in [Chapter 47](#), whilst [Chapter 15](#) provides more information on the mode of action and microbial resistance to radiation). One advantage of irradiation is that it does not cause a significant rise in temperature and hence has been called 'cold' sterilization. The main target site for radiation is DNA, but damage to other vital components such as RNA, enzymes and cell membranes is also involved. Single-stranded or double-stranded DNA breaks will inhibit DNA synthesis or cause errors in protein synthesis. Damage to the sugars and bases may also occur.

Ionizing radiation can be used to sterilize items that cannot be sterilized with heat (see [Table 17.1](#)). This is not a process that is normally carried out in a standard

pharmaceutical manufacturing facility; instead specialized plants are used.

D values are used in radiation sterilization (see [Fig. 16.1](#)) and can be calculated from the formula

$$D = \text{radiation dose} / (\log N_0 - \log N) \quad (16.2)$$

where N_0 and N represent a 1-log difference in numbers.

As with heat sterilization, the dose—response curve to radiation can vary (shoulders and tailing off can occur) but in general the exponential model holds. The initial lag that causes a shoulder is thought to result from multiple targets being hit before death or from DNA repair taking place (see [Chapter 17](#)).

Despite having been shown to have activity against bacterial spores, viruses and vegetative cells, UV radiation is not used for sterilization of pharmaceutical products because of its low penetrative power and absorption by glass and plastics. It is used for disinfection of surfaces, including isolators and safety cabinets, and can be used as part of the treatment for drinking water.

Filtration sterilization

Thermolabile solutions can be sterilized by filtration through filters that remove bacteria. Filtration sterilization is the complete removal of microorganisms within a specific size range from liquids or gases. It is a nonterminal sterilization process, and strict aseptic techniques need to be observed. Because of their small size, viruses are not removed by sterilization filtration and therefore, where possible, terminal sterilization processes should be preferred. Items that might be filter sterilized include heat-sensitive injections and ophthalmic solutions, biological products and air and other gases for supply to aseptic areas.

There are two main types of filtration mechanisms:

- Sieving, which makes use of synthetic membrane filters. This is an absolute mechanism since it ensures the exclusion of all particles above a defined size.
- Adsorption and trapping, which make use of depth filters. Depth filters have a high dirt-handling capacity and are often used as prefilters.

Generally, only membrane filters are thought to be suitable for the removal of microorganisms. The types used for pharmaceutical applications are usually made from cellulose esters or other polymers and are highly uniform with regular spaces or holes. When a liquid passes through the filter, all particles and microorganisms larger than the holes are retained. Membrane filters have the advantage of removing particles and microorganisms efficiently while retaining very little of the product in their holder or housing. However, they can become blocked quickly if the liquid being sterilized contains a lot of particles, and smaller particles can become trapped in the holes, causing a pressure differential. Because of this, prefilters are often used.

High-level disinfection

The term 'high-level' disinfection is often employed with chemical biocides that have demonstrated sporicidal activity. They are sometimes described as 'sterilants' or 'chemosterilants'. High-level disinfectants are usually considered to be highly reactive against macromolecules and are generally divided into two main groups of highly reactive biocides: alkylating and oxidizing agents (Table 16.2). Among the former, the aldehydes, notably formaldehyde (gas), glutaraldehyde and *ortho*-phthalaldehyde (OPA) are the most important. Oxidizing agents are composed of a wider family, such as peroxygen compounds (e.g. peracetic acid and hydrogen peroxide), chlorine dioxide and superoxidized water. Peracetic acid has also been used for the cold sterilization of some pharmaceutical preparations such as emulsions, hydrogels, ointments and powders.

Oxidizing agents have been shown to react extensively with macromolecules such as amino acids, peptides, proteins and lipids, notably through a reaction with sulfhydryl (–SH) groups, sulphur bonds (S–S) and fatty acid double bonds; however, the mechanisms by which they are lethal to the bacterial cell have not been resolved, although there is evidence that the primary target site for oxidizing agents used at a high (in use) concentration is the bacterial nucleic acid. Oxidizing agents also lead to the formation of peroxy radicals (fatty acids, lipids), hydroxyl radicals and other reactive species, although the role of these species in the lethality of these agents at a high concentration is subject to debate.

Alkylating agents have been shown to react with macromolecules through a reaction with amino, carboxyl, thiol, hydroxyl and imino groups and amide substituents. Their interactions with proteins and enzymes and their cross-linking ability are thought to be responsible for their mechanisms of

bactericidal action, but have also excluded them from British guidelines for endoscope disinfection (British Society of Gastroenterology, 2017), due to concerns centred on the potential fixation of prions. The lipophilicity of OPA is thought to be responsible for a better penetration of the aldehyde through the cell wall and for an increase in activity compared with glutaraldehyde.

On occasion, the use of chemical biocides has been combined with elevated temperatures (e.g. LTSF as discussed earlier in this chapter). For the disinfection of medical devices such as endoscopes, high-level disinfectants are often used in automated washer disinfectors, which are designed for specific devices and ensure automation of the process, although a precleaning stage is often necessary before disinfection takes place.

New technologies

Although only five sterilization procedures are usually recommended in pharmacopoeias, there has been an interest in developing alternative methods to overcome the disadvantages of existing ones (see Chapter 17). Most of the progress has been made in the food area. These technologies include the use of ultrahigh pressure, high-intensity light pulses, ultrasonication and gas plasma, the last being the most promising for the sterilization of medical devices and products. The principles of these new sterilization processes, although they are not used for pharmaceutical preparations, are worth mentioning briefly.

Ultrahigh pressure

The principle of using high pressure is that vegetative microorganisms are inactivated at pressures above 100 MPa

Table 16.2 Sterilization and high-level disinfection using chemical biocides

Chemical agents	Use
Hydrogen peroxide	Gas plasma sterilization (endoscopes)
Peracetic acid	Endoscopes, pharmaceutical preparations
Chlorine dioxide	Gas-phase chlorine dioxide sterilization (medical equipment)
Glutaraldehyde	High-level disinfection (endoscopes)
<i>o</i> -Phthalaldehyde	High-level disinfection (endoscopes)
Formaldehyde	Gaseous sterilization (LTSF)
Ethylene oxide	Liquid and gaseous sterilization
LTSF, low-temperature steam formaldehyde.	

and bacterial spores are inactivated at pressures above 1200 MPa. The use of high-pressure processing for food preservation has been combined with the chemical effect of the preservative system, such as a low pH in certain foodstuffs (e.g. jams, fruit juices). The low pH ensures the prevention of outgrowth of bacterial spores. The advantage of this process is that the quality and taste of the products tend not to be affected by such a system. Indeed, high pressure tends to preferentially denature macromolecules rather than low molecular weight flavour and odour compounds (Yordanov & Angelova, 2010). The microbial cell offers multiple pressure-sensitive target sites (e.g. enzymes, membranes, genomic material), although bacterial spores are more resilient to high pressures. The combination of high pressure and elevated temperature has been shown to be synergistic, although the process varies depending on the combination and the type of spores.

The use of high pressure for the sterilization of pharmaceutical products has been considered and might offer an appropriate alternative to existing sterilization processes. High-pressure sterilization is an innocuous process that can be performed at a relatively low temperature. Another advantage of using high pressure is the rapid control of thermal processes. Indeed, the application of pressure raises the temperature but conversely the temperature decreases as the pressure is reduced. Hence rapid temperature changes following a rapid change in pressure can help in controlling a thermal process and reduce heat-induced damage to a product. However, detailed studies with pharmaceutical preparations and detailed process validation are needed for high pressure to be used for the sterilization of dosage forms.

High-intensity light pulses

The application of intense light, such as from a high-intensity laser, is known to inactivate microorganisms. This principle already has applications in the food industry and also in the medical area, notably in dentistry. Such processes have been shown to inactivate vegetative microorganisms and bacterial spores. In the food industry, broad-spectrum light with pulse durations from 10^{-6} to 10^{-1} seconds and with energy densities ranging from 0.1 J cm^{-2} to 50 J cm^{-2} is used. In dentistry, high-intensity light pulses (i.e. from a laser) have been combined with the use of antimicrobial dyes (e.g. toluidine blue) to achieve better inactivation of microorganisms in the treatment of root canal infections, and such a combination is also referred to as antimicrobial photodynamic treatment.

For pharmaceutical preparations, a potential application is the terminal sterilization of clear solutions such as water, saline, dextrose and ophthalmic products. The type of

container is of prime importance since it must not hinder the transmission of light.

Ultrasonication

The use of sonication to inactivate microorganisms was first reported more than 30 years ago. The principle is based on cavitation through the material exposed, resulting in the formation and collapse of small bubbles. The ensuing shock waves associated with high temperatures and pressures can be sufficiently intense to disrupt the microbial cell; however, spores are highly resistant. A synergistic effect has been reported by combination of ultrasound and heat to a certain extent. Such a combination has been reported to reduce the heat resistance of microorganisms.

Gas plasma

Gas plasma is generated with the application of a strong magnetic field to a gas-phase compound (e.g. hydrogen peroxide; H_2O_2). This process creates a mixture of charged nuclei, free electrons and other reactive species such as free radicals that can then damage cellular components (e.g. membrane, nucleic acid), a mechanism of action similar to that of oxidizing agents. Plasma is generally considered the 'fourth' state of matter. This dry sterilization process is effective against vegetative microorganisms and also bacterial spores and has the advantage of being a nonthermal method. The use of gas plasma offers an appropriate alternative to traditional sterilization processes and finds many applications for the sterilization of medical devices, but also in drug delivery with the treatment of biomaterials (Fig. 16.4).

New healthcare arena technologies

In recent years there has been a move towards low temperature sterilization; this has been driven by a number of market factors including reduced cost, user friendliness and improved material compatibility.

A combination of H_2O_2 vapour and gas plasma can be used for both heat- and moisture-sensitive medical instruments, with technologies combining ozone and H_2O_2 vapour having shown increased antimicrobial efficacy against a range of microorganisms.

In 2016, the FDA approved the use of a nitrogen dioxide (NO_2) technology for sterilization purposes; NO_2 is compatible with most polymers used in medical devices and demonstrates positive results against spores. It is particularly useful for products, such as prefilled syringes and drug-delivery systems, which may be radiation, heat and moisture sensitive (Food and Drug Administration, 2016).



Fig. 16.4 Gas plasma sterilizer from Advanced Sterilization Products.

Summary

Sterilization is an essential process for the manufacture of sterile dosage forms, and reprocessing medical devices and products. Several processes can be used to achieve appropriate sterilization for a given preparation or product/device. Each of these processes has advantages and disadvantages, although steam sterilization remains the reference standard. The advances in nonthermal sterilization and the coming new technologies, although mainly applied in the food industry to date, offer potentially valuable alternatives. The demonstration of sporicidal activity of a new technology, as well as its control and reproducibility, remains essential.

Common to all these processes is the need for the user to understand the technology, its activity and limitations, to follow the appropriate guidelines but also, importantly, to ensure the validation of the process. Failure to provide the appropriate documentation and to control a sterilization process adequately might result in failure of the process, with potentially fatal consequences.

It is important to note at this point that although terminal sterilization ensures the destruction of possible microbial contaminants, it needs to be operated alongside Good Manufacturing Practice (Chapter 54). Therefore suitable measures must be taken to ensure the microbiological quality of pharmaceutical preparations during manufacture but also during packaging, storage and distribution. These important aspects are discussed in more depth in Chapter 17.

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Questions

1. Sterilizing grade filters:
 - A. are often made of polymeric membranes
 - B. usually remove all microorganisms including viruses
 - C. can be used for terminal sterilization
 - D. are the preferred sterilization method in pharmacopoeias
 - E. cannot be used to sterilize gases
2. The *D* value:
 - A. is the time taken to kill all organisms present
 - B. is used to calculate the efficacy of filtration sterilization
 - C. is the time taken to achieve a one-log reduction in microorganisms
 - D. is the time taken to kill 99% of the microorganisms present
 - E. is used if the destruction curve follows the nonlinear logarithmic model
3. Heat sterilization:
 - A. can be accomplished by boiling items for 10 minutes
 - B. is effective against spongiform encephalopathies
 - C. usually follows a first-order reaction kinetic
 - D. is more effective against bacterial endospores than vegetative cells
 - E. is not effective against viruses
4. Pharmacopoeias recommend which of the following methods for the sterilization of pharmaceutical products:
 - A. hydrogen peroxide
 - B. *o*-phthalaldehyde
 - C. formaldehyde fumigation
 - D. ethylene oxide gas
 - E. 70% industrial methylated spirits
5. Steam sterilization:
 - A. is most effective when wet saturated steam is used
 - B. results in a rapid rise in temperature because of the transfer of latent heat
 - C. utilizes a hot air oven
 - D. is most effective when superheated steam is used
 - E. is usually slower than dry heat sterilization
6. Dry heat sterilization:
 - A. is carried out in an autoclave
 - B. generally has shorter hold times than steam sterilization
 - C. is usually carried out under high pressure
 - D. kills microorganisms via oxidation
 - E. is not able to depyrogenate glassware
7. Oxidising gases:
 - A. include hydrogen peroxide and ozone
 - B. are relatively stable
 - C. are a recommended pharmacopoeial method of sterilization
 - D. include ethylene oxide
 - E. usually have good penetrating power
8. Radiation sterilization of pharmaceutical products:
 - A. can be accomplished using ultraviolet light
 - B. is normally carried out in a standard pharmaceutical manufacturing facility
 - C. causes a significant rise in temperature
 - D. can only be used to sterilize the surface of items as it is nonpenetrating
 - E. utilizes γ -rays or high-speed electron beams
9. Low temperature steam formaldehyde:
 - A. is easy and safe to use as it is nontoxic to humans
 - B. is usually carried out at 121 °C
 - C. utilizes steam below atmospheric pressure
 - D. is a recommended pharmacopoeial method of sterilization
 - E. can never be used on medical devices
10. Ethylene oxide:
 - A. is not a terminal sterilization method
 - B. can permeate through many polymeric materials
 - C. has a boiling point of 70 °C
 - D. is not effective against protozoa
 - E. is more commonly used in hospitals than pharmaceutical manufacturing

Sterilization in practice

Jean-Yves Maillard, Katie Laird, Susannah E. Walsh

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KEY POINTS

- Sterilization is essential to produce sterile dosage forms.
- Pharmacopoeias usually recognize four terminal sterilization processes – steam (under pressure), dry heat, gaseous (ethylene oxide) and ionizing radiation sterilization – and nonterminal sterilization by filtration.
- The choice of sterilization processes reflects the great diversity of pharmaceutical preparations, medicinal products and devices that are required to be sterile.

- A sterilization process is informed by a number of specific standard documents and guidelines, is tightly controlled and must be documented and validated.
- A clear understanding of the method, the product to be sterilized (including its packaging), the validation process and the overall documentation required is necessary to perform a successful sterilization.

Sterile products

Sterilization is an essential part of the processing of pharmaceutical dosage forms that are required to be sterile. By definition, a sterile product is completely free of viable microorganisms. In addition to the pharmaceutical products that need to be sterile, a number of medical devices that come into contact with sterile parts of the body or are reused in patients also need to be free of microorganisms (Table 17.1). The diversity of the items to be sterilized in terms of properties (e.g. heat sensitive or not), bulk, content and the number of items to be sterilized per load requires the use of distinct sterilization processes. The *British Pharmacopoeia* ([British Pharmacopoeia Commission, 2020a](#)), as an example, describes the use of five main sterilization processes to accommodate the range of products to be sterilized: steam, dry heat, gaseous, ionizing radiation and filtration sterilization. The first four methods are usually used to process products in their final containers (terminal sterilization). Regardless of the sterilization method used, it is important that the process itself is fully validated. A number of guidelines and European/international standard documents for specific product–sterilization method combinations exist and are followed by manufacturers and end users. Failure to control and/or document adequately a sterilization process can lead to serious incidents. This chapter aims to provide a brief overview of the recommended sterilization processes, their control and their validation.

Table 17.1 Examples of sterile preparations and devices

Preparation/product/item	Typical volume	Typical container	Sterilization process
Injections			
Intravenous infusion, e.g. blood products	0.5 L	Plastic, glass	Steam Filtration (e.g. addition of additives)
Parenteral nutrition fluid	>3 L	Plastic, glass	Steam Filtration (e.g. addition of vitamins)
Small-volume injections, e.g. insulin, vaccines	1 mL to 50 mL	Plastic glass	Steam ^a Filtration
Small-volume oily injections		Glass	Dry heat
Noninjectable sterile fluids			
Noninjectable water, e.g. surgery, irrigation	0.5 L to 1 L	Plastic (polyethylene or polypropylene)	Steam
Urological irrigation solution	>3 L	Plastic (rigid)	Steam Filtration
Peritoneal dialysis and haemodialysis solutions	2.5 L	Plastic	Steam
Nebulizer solutions	2.5 mL	Plastic (polyethylene nebulers)	Steam Filtration
Ophthalmic preparations			
Eye drops	0.3 mL to 0.5 mL	Plastic, glass	Steam ^a Filtration
Eye lotions	>0.1 L	Plastic, glass	Steam
Eye ointments	—	Plastic, aluminium	Dry heat Filtration
Contact lens solutions	Small	Plastic	Chemical disinfection
Dressings			
Chlorhexidine gauze dressing		Different wrapping ^b	Steam ^c
Polyurethane foam dressing			Dry heat
Elastic adhesive dressing			Ethylene oxide
Plastic wound dressings			Ionizing radiation Other effective method
Implants			
		Small, sterile cylinders of drug	Dry heat Chemical (0.02% phenyl mercuric nitrate, 12 hours, 75 °C)
Absorbable haemostats			
Oxidized cellulose, human fibrin foam			Dry heat

Continued

Table 17.1 Examples of sterile preparations and devices—cont'd

Preparation/product/item	Typical volume	Typical container	Sterilization process
Surgical ligatures and sutures			
Sterilized surgical catgut			γ-radiation Chemical (96% ethanol + 0.002% phenyl mercuric nitrate + formaldehyde in ethanol 24 hours before use; naphthalene or toluene at 160 °C for 2 hours)
Nonabsorbable type			γ-radiation Steam
Instruments and equipment			
Syringes		Glass, plastic	Dry heat Steam γ-radiation Ethylene oxide
Metal instruments			Steam
Rubber gloves			γ-radiation Ethylene oxide
Respirator parts			Steam
Fragile heat-sensitive devices			Chemical disinfection
^a Depends on the preparations (thermostable or thermolabile). ^b Dressings must be appropriately wrapped (aseptic handling) for their specific use. ^c Sterilization process depends on the stability of the dressing constituents (e.g. dressings containing waxes cannot be sterilized by moist heat) and the nature of their components.			

Determination of sterilization protocols

Various technologies are available to achieve sterility of pharmaceutical preparations and medical devices (Table 17.2). Generally, sterilization of the product in its final container (terminal sterilization) is preferred. This implies that the container must not impinge on the optimum sterilization to be delivered and that the container and closure maintain the sterility of the product throughout its shelf life. The selected sterilization process must be suitable for its purpose, i.e. the sterilization of a given product, device and preparation, which means that the product and its container have to be rendered sterile and must not be damaged by the process.

The choice of an appropriate sterilization process depends on a number of factors (Table 17.3) related to the product to be sterilized, such as the type and composition of product and also the quantity to be sterilized. Additionally, the composition and the packaging of the product are significant factors that rule out some sterilization processes. For example, a heat-labile preparation would not be sterilized by heat sterilization, and a small oily injection would not be sterilized by steam sterilization (further examples are given in Table 17.1). For specific types of products such as dressings, although moist heat sterilization is generally the method of choice, only certain types of autoclave, such as vacuum and pressure-pulsing autoclaves, are appropriate.

For any given preparation or product, it is difficult to predict the microbial bioburden prior to sterilization. It is

Table 17.2 Sterilization technologies (for pharmaceutical preparations and medical devices)

Type	Principle	Examples
Terminal sterilization		
Physical	Heat	Steam Dry heat
	Radiation	γ-radiation Accelerated electrons (particle radiation)
Chemical	Gaseous	Ethylene oxide Low-temperature steam formaldehyde Gas plasma Peracetic acid, hydrogen peroxide
Nonterminal sterilization		
Filtration	Aseptic procedure	

assumed that the bioburden of pharmaceutical preparations will be minimal as the manufacturing process should adhere to Good Manufacturing Practice (GMP) (Box 17.1 and Chapter 54). However, a sterilization process should be able to deal with a worst-case scenario. This is usually exemplified by the use of biological indicators (see the section entitled 'Process indicators' later in this chapter) such as bacterial spores, which are considered the most resistant microorganisms (with the exception of prions, the agents responsible for spongiform encephalopathies). This is usually the situation for official sterilization methods. Pharmacopoeial recommendations, as well as regulatory guideline documents, are derived from data generated from the use of biological indicators for a given sterilization process.

When a fully validated sterilization process has been conducted, the release of a batch of product can be based on process data obtained during sterilization rather than the results from sterility testing. Any change in the sterilization procedure (e.g. product load, type of containers) requires revalidation to take place.

Resterilization of products/devices can cause their degradation (e.g. repeated irradiation or autoclaving) or may even cause them to become toxic (e.g. with ethylene oxide; Richards, 2004). Therefore, any proposed rest-erilization must be carefully investigated.

Recommended pharmacopoeial sterilization processes

Five main sterilization processes, which have different characteristics, are usually recommended by pharmacopoeias:

- steam (under pressure) sterilization (terminal);
- dry heat sterilization (terminal);
- ionizing radiation sterilization (terminal);
- gaseous (ethylene oxide) sterilization (terminal); and
- sterilization by filtration (nonterminal).

Although the use of other sterilization methods is not necessarily precluded, appropriate validation documentation for each product needs to be provided. More information can be found in Chapters 15 and 16, or in the relevant pharmacopoeia. At the time of writing, examples of these include the *European Pharmacopoeia* (European Pharmacopoeia Commission, 2020), the *United States Pharmacopoeia* (United States Pharmacopoeial Convention, 2020) and the *British Pharmacopoeia* (British Pharmacopoeia Commission, 2020a) but it is always important to consult the most up-to-date texts and guidelines. The European Medicines Agency (EMA) has published a *Guideline on the Sterilisation of the Medicinal Product, Active Substance, Excipient and Primary Containers*, which includes decision trees for the selection of sterilization methods (European Medicines Agency, 2019).

Steam (under pressure) sterilization

Steam sterilization is the most reliable, versatile and universally used form of sterilization and relies on the combination of steam, temperature and pressure. The typical cycle consists of a holding time of 15 minutes at a temperature of 121 °C at 15 psi (103 kPa) gauge pressure (Table 17.4). The aim is to deliver steam at the phase boundary (dry saturated steam; see Fig. 16.2) to all areas of

Table 17.3 Selection of a sterilization process

Type of product/preparation		
Pharmaceutical preparations	Volume Composition	Large, small injection Water, oil, powder
Medical devices	Size Composition	Small, large, complex devices (e.g. endoscopes, respirator parts) Plastic, glass, metal, porous (e.g. dressing)
Possible damage to the preparation/product		
	Heat (heat-sensitive preparations) Radiation (water) Corrosiveness (oxidizing agents)	
Possible damage to the product/container		
	Water ballasting Moisture Glass breaking (on cooling) Change in composition (irradiation) Corrosiveness	
Other considerations		
Toxicity/safety	Gas sterilization (ethylene oxide, formaldehyde) Liquid sterilants: aldehydes Radiation sterilization: radioactive source Steam sterilization: steam water quality	
Level of bioburden	Expected heavy contamination Surgical instruments	
Sterilization regimen	Local sterilization (portable autoclave) Large quantity of items to be sterilized Need for quarantine (desorption of toxic chemicals)	
Cost of sterilization process	Equipment, e.g. autoclave, electron accelerator Facility, e.g. irradiation plant Running cost: gas, ⁶⁰ Co Training of end users	
Validation	Ease of validation; producing appropriate documentation Cost of audit Cost of validation	

Box 17.1 Key points to achieve Good Manufacturing Practice for production of sterile pharmaceutical products

- Qualified personnel with appropriate training
- Adequate premises
- Suitable production equipment, designed for easy cleaning and sterilization
- Adequate precautions to minimize the bioburden before sterilization (starting materials, etc.)
- Validated procedures for all critical production steps
- Environmental monitoring and in-process testing procedures

the load. This is achieved using steam and pressure (Table 17.5).

Steam under pressure is commonly used unless prohibited by lack of load penetration or heat and/or moisture damage. Steam can only kill microorganisms if it makes direct contact with them, so it is very important to avoid air pockets in the sterilizer during a sterilization process. In addition, air can reduce the partial pressure of the steam such that the temperature reached on surfaces will be less than that expected with the pressure used. Hence removal

Table 17.4 Typical terminal sterilization cycles

Sterilization process	Temperature (°C)	Pressure (psig)	Relative humidity (%)	Holding time/dose	Concentration	Parametric release	Desorption
Heat							
Moist heat	121	15 (103 kPa)	—	15 min	—	Yes	No
	134	30 (207 kPa)	—	3 min	—	Yes	No
Dry heat	160	—	—	>2 h	—	Yes	No
Radiation							
γ-radiation	Room	—	—	25 kGy ^a	—	Yes	No
Particle radiation	Room	—	—	25 kGy	—	No	No
Gaseous							
Ethylene oxide ^b	40–50	—	40–80	30 min to 10 h	400 mg L ⁻¹ to 1000 mg L ⁻¹	No	Yes ^c
LTSF ^d	70–80 ^e	—	75–100	90 min	6 mg L ⁻¹ to 50 mg L ⁻¹	No	Yes

^aStandard dose. The time necessary to achieve this dose depends on the source. For γ-ray irradiation, the process can take up to 20 hours, whereas for high-energy electrons (particle radiation), only a few minutes may be required.

^bVacuum cycle; pretreatment of the load: preheating and humidification of the load. Pressurized cycle: always higher than atmospheric pressure; allows shorter contact time.

^cDesorption could take up to 15 days; maximum threshold of ethylene oxide residues and evaluation documented in ISO 10993-7 ([International Organization for Standardization, 2008](#)).

^dValues can differ slightly depending on the literature.

^eLower temperature of 55 °C to 56 °C can be used depending on the thermotolerance of the preparation.

LTSE, low-temperature steam formaldehyde.

Table 17.5 Examples of temperature and pressure combinations used for steam sterilization

Temperature (°C)	STEAM PRESSURE	
	kPa	psig
115	69	10
121	103	15
126	138	20
134	207	30

Steam pressures are expressed in kilopascals (kPa) and pounds per square inch gauge (psig), the latter still finding continuing use.

of air is an essential part of the process to ensure effective sterilization. To remove the air present when an autoclave is loaded, autoclaves are equipped with air removal/displacement systems (e.g. vacuum and displacement autoclaves). For porous loads, gravity displacement systems (downward-displacement autoclaves) are not adequate, and vacuum and pressure-pulsing autoclaves are preferred ([McDonnell, 2007](#)). Noncondensable gases must also be removed and monitored; these are atmospheric gases such as nitrogen and oxygen that form part of the initial atmosphere of the sterilizer. Other factors that affect the efficacy of steam sterilization are water content and steam purity. The optimal sterilization is obtained with saturated steam (as discussed in [Chapter 16](#)). Supersaturated steam (i.e.

wetter steam) is associated with condensation and poor penetration. Superheated steam (i.e. drier steam) behaves like dry heat and is less efficient. Steam purity is determined by the quality of the water, which can be affected by a number of contaminants (e.g. pyrogens, amines, toxic metals, iron, chlorides) that can render the sterile product unsafe (e.g. toxicity caused by pyrogenic reactions, metallic poisoning) or damaged (e.g. discoloration of packaging, corrosion caused by iron and chlorides).

Steam under pressure is generated in autoclaves which can vary greatly in size and shape, ranging from portable bench-top units to industrial production facilities (Fig. 17.1). A cross-section through an autoclave is shown in Fig. 17.2.

Steam sterilization applications are informed/regulated by a number of European and international guidelines and standards providing information on sterilizer design and installation, quality of steam, requirement for pressure, development and validation and routine control, etc.



Fig. 17.1 Examples of autoclaves. **(a)** Square section, **(b)** Square section during loading, **(c)** Swiftlock Compact (loading) and **(d)** Swiftlock Compact (closed). (Courtesy of Astelle.)

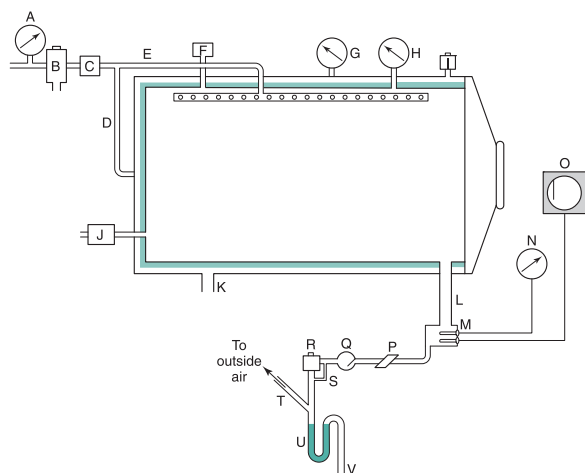


Fig. 17.2 The features of a large steam sterilizer (for simplicity, the control valves have been omitted). A, mains pressure gauge; B, separator; C, reducing valve; D, steam supply to jacket; E, steam supply to chambers; F, air filter; G, jacket pressure gauge; H, chamber pressure gauge; I, jacket air vent; J, vacuum pump; K, jacket discharge channel (detail not shown); L, chamber discharge channel; M, thermometer pocket; N, direct-reading thermometer; O, recording thermometer; P, strainer; Q, check valve; R, balanced-pressure thermostatic trap; S, bypass; T, vapour escape line; U, water seal; V, air-break.

Dry heat sterilization

The most common dry heat sterilization method uses hot air ovens (Fig. 17.3). Other procedures, such as sterilizing tunnels using high-temperature filtered laminar air flow or infrared irradiation to achieve rapid heat transfer, are also available. Hot air ovens are usually heated electrically and often have heaters under a perforated bottom plate to provide convection currents (gravity convection type). Mechanical convection hot air ovens are equipped with a fan to assist air circulation and increase heat transfer by convection (Joslyn, 2001). Dry heat sterilization is less expensive than steam sterilization and is effective for the depyrogenation of containers/packaging (e.g. glassware). Overloading should be avoided, wrappings and other barriers minimized and the load positioned to allow optimal air circulation. Other problems include long heating up times (e.g. with large loads of instruments) and the charring or baking of organic matter onto items. Dry heat sterilization cycles are generally longer than for moist heat sterilization, typically 2 hours at 160 °C (see Table 17.4). The process is thermostatically controlled and monitored using thermocouples.

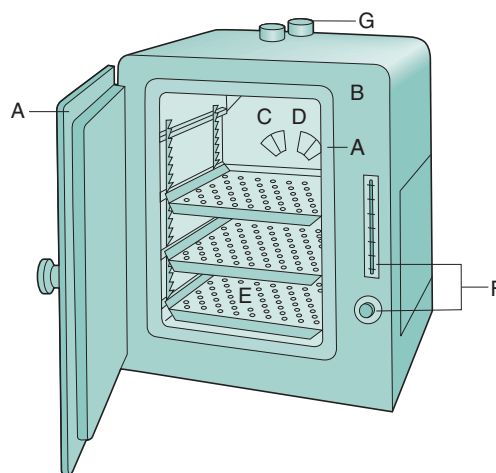


Fig. 17.3 Hot air oven. A, heat-resistant gasket; B, outer case containing glass-fibre insulation, and heaters in chamber wall; C, false wall; D, fan; E, perforated shelf; F, regulator; G, vents.

Dry heat sterilization cannot be used for a number of products, such as rubber, plastics and other thermolabile items, or for aqueous solutions.

Integrated lethality in sterilization practice

All heat sterilization processes must include heating up and cooling down periods. These prolonged periods at a raised temperature may increase the degradation of the product. Integrated lethality attempts to examine the effects of heat on the inactivation process during these periods.

For steam sterilization, the F_0 concept ('reference unit of lethality') is used. This takes into account the heating up and cooling down stages of the heating cycle and is expressed as the equivalent time in minutes at a temperature of 121 °C delivered by the process to the product in its final container with reference to microorganisms with a Z value of 10. Its calculation is complex, and further information can be found in the relevant pharmacopoeias. In practice, computer programs can be used to calculate the combined effect of whole processes, allowing a reduction in the total process time. It is important that the appropriate sterility assurance level is consistently achieved, and the routine use of biological indicators is recommended, although following process validation, parametric release might be preferred.

Gaseous sterilization

The gaseous sterilization method recommended by pharmacopoeias mainly employs the alkylating agents (in particular ethylene oxide), although oxidizing agents (hydrogen peroxide and peracetic acid) are also mentioned. It is usually used on a commercial scale for the sterilization of catheters, infusion giving sets, syringes, prostheses and some plastic containers and thermolabile powders (if humidity is not a problem). The ethylene oxide sterilization cycle is complex since many factors need to be controlled over a long period (Table 17.4). The control of the temperature, concentration and relative humidity is critical. In addition, ethylene oxide is very flammable and can form explosive mixtures in air. It is therefore combined with an inert gas carrier (e.g. carbon dioxide or nitrogen). Ethylene oxide is also toxic, mutagenic and a possible human carcinogen. Gaseous sterilization using ethylene oxide is nevertheless a popular sterilization process, mainly because of the low temperature used during sterilization, but also because of the amount of information acquired on ethylene oxide sterilization processes over the years. The sterilization procedure is usually carried out in a purpose-built, gas-tight stainless steel chamber which can withstand high pressures and a high vacuum. However, systems using a slight negative pressure rather than drawing a full vacuum are available (Fig. 17.4) and these are suitable for smaller, vacuum-sensitive loads.

Packaging should be permeable by air, water vapour and ethylene oxide. The sterilized products need to be quarantined after the process to allow the removal of gas. The *European Pharmacopoeia* and other international standards set limits for ethylene oxide residue levels (e.g. a maximum of 10 ppm for plastic syringes).

Low-temperature steam formaldehyde (LTSF, discussed in Chapter 16), although not included in this chapter's list of recommended methods, is used for the sterilization of certain preparations. As with ethylene oxide, its sterilization cycle is rather complex as several parameters have to be controlled (see Table 17.4).

Radiation sterilization

There are two types of radiation unit. The becquerel (Bq) measures the activity of a source of radiation (physical radiation). One Bq equates to a source that has one nuclear disintegration per second. The gray (Gy) measures the effect of radiation on living tissue. One Gy is equal to the transfer of 1 J of energy to 1 kg of living tissue. The gray has replaced the rad that quantified radiation absorbed dose. The electronvolt (eV) measures the energy of radiation and is usually expressed as millions of electronvolts (MeV).

The source of γ -rays for sterilization is usually cobalt-60. Caesium-137 can also be used but has less penetrating power. Cobalt-60 decays with the emission of two high-energy γ -rays (1.17 MeV and 1.33 MeV) and a



Fig. 17.4 Examples of ethylene oxide sterilizers using slight negative pressure rather than the conventional vacuum system. These are suitable for smaller loads, e.g. hospital reprocessing loads, research and development work, short production runs and low-volume production. (Courtesy of Andersen Caledonia.)

lower-energy (0.318 MeV) β -particle. Gamma radiation is highly penetrative, causes negligible heating of the sterilized product at normal doses and induces no radioactivity in the final product.

Irradiation of a product can be carried out in batches but is more commonly a continuous process using a conveyor system. The products pass through the irradiation chamber and are irradiated from one or two sides. The source is shielded with concrete to protect the operators and the environment. The intensity of radiation decreases as it penetrates. For example, 100 mm of a product with a density of 1 g cm^{-3} would reduce the cobalt-60 intensity by 50%. A cobalt-60 source of $1 \times 10^{16} \text{ Bq}$ to $4 \times 10^{16} \text{ Bq}$ is used for industrial irradiation, and this provides a radiation dose in excess of 25 kGy. In most of Europe, 25 kGy is the standard dose (e.g. [European Pharmacopoeia Commission, 2020](#)). When not in use, the radioactive source is submerged in water for shielding and cooling.

Particle radiation sterilization uses β -particles that are accelerated to a high energy by application of high-voltage potentials (no radioactivity required). Their low energy means that beams from particle accelerators are less penetrating than γ -rays, with only 10 mm of a 1 g cm^{-3} material being penetrated per million electronvolts (MeV). However, an important advantage of particle radiation sterilization is that the source can be turned off and is directional ([Lambert, 2013](#)). The design of an accelerator can be customized to particular applications by including different energy and power requirements. The beam source is shielded with concrete and products are conveyed through the exposure area and irradiated. Another advantage is that shorter exposure times are required than those needed for γ -ray irradiation. High-energy beams with energies of 5 MeV to 10 MeV are used for sterilization, the accelerating field being generated using radiofrequency or microwave energy. Once it has been accelerated to the required energy, the beam of electrons is controlled by magnetic fields which can alter its size, shape or direction ([McDonnell, 2007](#)).

Radiation can affect a number of materials (e.g. polyethylene, silicone rubber, polypropylene, Teflon), aqueous solutions (e.g. through the process of water radiolysis) and packaging (discussed further in the 'Limitation of sterilization methods' section later in this chapter). Although radiation sterilization is considered a 'cold' process, intense radiation can cause an increase in temperature and as such possible overheating needs to be considered for a specific load.

Validation of radiation sterilization involves the use of *Bacillus pumilus* as a biological indicator and dosimetric analysis (discussed later in this chapter). The routine monitoring involves measurements to ensure that all products are receiving the required dose. The radiation sterilization procedure is highly regulated, and there are a

number of European and international standards and guidelines available with information on requirements for the development, validation and routine control of the process (e.g. BS EN ISO 11137-1) and the dose required for sterilization (e.g. ISO 11137-2).

Filtration

Filtration is employed for nonterminal sterilization and has to be used under strict aseptic conditions. It is used for those preparations that cannot be sterilized by a terminal process or to which an agent (e.g. excipient, heparin, vitamin) is added post-sterilization. Filtration is used to sterilize aqueous liquid, oils and organic solutions, and also air and other gases. Membrane filtration is an absolute process which ensures the exclusion of all particles larger than a defined size. Although many materials have been used to make filters, only a few are suitable for sterilization of pharmaceutical products.

Depth and surface filters are suitable for prefiltration of pharmaceutical products as they can retain large amounts of particles. Depth filters can be made of fibrous, granular or sintered material that is bonded into a maze of channels that trap particles throughout their depth. Surface filters are made of multiple layers of a substance such as glass or polymeric microfibres. Any particles that are larger than the spaces between the fibres are retained, and smaller particles may be trapped in the matrix ([McDonnell, 2007](#)). A membrane filter downstream is needed to retain any fibres shed from these filters, as well as small particles and microorganisms.

To sterilize a product, it is often necessary to combine several types of filtration (e.g. depth, surface and membrane filters) to achieve the removal of microorganisms. Depth and surface filtration are used to remove the majority of the particles by acting as prefilters. The final filtration step is accomplished using a membrane filter. This combined approach removes particles and microorganisms without the membrane filter being rapidly blocked up with large particles.

High-level disinfection

In addition to the processes previously described, high-level disinfectants (chemical biocides) have to be mentioned as they are used for the *chemosterilization* of medical devices, particularly high-risk items that come into contact with sterile parts of the body, such as surgical instruments, intrauterine devices and endoscopes (which are used for a wide range of diagnostic and therapeutic procedures) ([Table 17.1](#)).

Like the gaseous biocides, the activity of high-level liquid disinfectants depends on a number of factors (Maillard & McDonnell, 2012). Consequently, the training of the end user is of vital importance. Guidelines are often available from professional societies regarding the use of chemical biocides and specific devices; for example, the sterilization procedure and risk assessment for gastroscopes are published by the British Society of Gastroenterology (2017).

To ensure the efficacy of high-level disinfection, knowledge of the factors affecting efficacy, education of end users and compliance with manufacturers' instructions is essential (Maillard & McDonnell, 2012). The main advantage of using high-level disinfection is the low temperature used in processing medical devices. However, high-level disinfection might not give the same level of sterility assurance, and where possible, physical processing (e.g. steam sterilization) should be the method of choice.

The main disadvantages of high-level disinfection are exposure toxicity with regard to the end users, damage to materials and potential emerging microbial resistance; all high-level disinfectants are toxic at the concentration used. For example, there have been many reports of exposure toxicity from glutaraldehyde following endoscope reprocessing, and this has resulted in abandonment of the use of the dialdehyde in many countries. Aldehyde and alcohol disinfectants are fixatives that may make it difficult to remove protein and so they should not be used to disinfect endoscopes under current British guidelines (British Society of Gastroenterology, 2017). Damage to the materials following reprocessing can take the form of corrosion of metallic surfaces and increased rigidity of plastics. Problems associated with inappropriate high-level disinfection regimens, which resulted in microbial contamination, have been described since the 1990s. Reports have highlighted the potential for transmission of infection via medical devices and medical device reprocessors. These events are quite distinct from reports that microorganisms are becoming resistant to the in use concentrations of these disinfectants, including high-level ones (Maillard, 2010; Maillard et al., 2013).

Statistical considerations of sterility testing and sterility assurance level

The strict definition of sterility is the complete absence of viable microorganisms. In other words, after a successful

sterilization process, the number of microbial survivors should be zero. This is an absolute definition which cannot be guaranteed, especially from a microbial point of view. To ensure the absence of viable microorganisms requires that all viable microorganisms can be detected and cultured. When one looks at microbial inactivation following, for example, exposure to heat or radiation, the inactivation usually follows first-order kinetics (see Chapters 7, 15 and 16), although in practice microorganisms are inactivated at different rates, producing a deviation from linear inactivation. Thus assuring the complete elimination of microbial contaminants and thus sterility of the product cannot be *guaranteed* mathematically or practically.

Instead then of defining sterility in a strict microbiological sense, it is more appropriate to consider the likelihood of a preparation being free of microorganisms. This is best expressed as the probability of a product containing a surviving microorganism after a given sterilization process. Survival depends on the number and the type of microorganisms, soiling and the environmental conditions within the sterilizing equipment. The concept of a sterility assurance level (SAL) or microbial safety index provides a numerical value for the probability of survival of a single microorganism. The SAL is therefore the degree of assurance for a sterilizing process to render a population of products sterile. For pharmaceutical preparations a SAL of 10^{-6} or better is required. This equates to not more than one nonsterile item per million items/units of the final product. Practically, the lethality of a sterilization process and in particular the number of log cycles required need to be calculated.

The inactivation factor, which measures the reduction in the number of microorganisms (of a known D value; see Chapters 15 and 16) brought about by a defined sterilization process, can be calculated as follows:

$$IF = 10^{t/D} \quad (17.1)$$

where IF is the inactivation factor, t is the contact time (for heat or gaseous process) or radiation dose (for ionizing radiation) and D is the D value appropriate to the process employed. An example calculation is shown in Box 17.2.

Calculation of the IF is based on obtaining inactivation kinetics that follow a first-order process. In reality, this is not always the case. In the food industry, the calculation of the most probable effective dose (MPED) is preferred as it is independent of the slope of the survivor curve for the process. However, to establish an MPED that will achieve the required reduction in a number of microorganisms requires complex calculations.

Box 17.2 Worked example

Consider steam sterilization. For an initial bioburden of 10^4 spores of *Geobacillus stearothermophilus*, an inactivation factor of 10^{10} will be required to achieve a sterility assurance level of 10^{-6} .

G. stearothermophilus has a *D* value of 1.5 for steam sterilization.

Thus according to Eq. 17.1, a 15-minute sterilization process (i.e. holding time) at 121 °C will be required to achieve an inactivation factor of 10^{10} (i.e. $10^{15/1.5}$).

The process will therefore reduce the level of microorganisms by 10 log cycles.

Test for sterility of the product

Sterility testing assesses whether a sterilized pharmaceutical or medical product is free from viable microorganisms by incubating all or part of the product with a nutrient medium. Testing for sterility is a destructive process. For an item to be shown not to contain organisms, unfortunately it has to be destroyed. Due to the destructive nature of the test and the probabilities involved in sampling only a portion of a batch, it is only possible to say that no contaminating microorganisms have been found in the sample examined in the conditions of the test (British Pharmacopoeia Commission, 2020b). Thus the measurement of sterility relies on statistical probability. In other words, it is impossible to prove sterility since sampling may fail to select nonsterile containers, and culture techniques have limited sensitivity. In addition, not all types of microorganisms that might be present can be detected by conventional methods as not all microorganisms are affected by a sterilization process in the same way. It is possible that some may not be killed or removed. For example, a filter pore size of 0.22 µm is usually used for filtration sterilization, which means that smaller microorganisms such as viruses are allowed to pass through.

Detailed sampling and testing procedures are given in pharmacopoeias, and further details can be found in Chapter 14. For terminally sterilized products, biologically based and automatically documented physical proofs that show correct treatment during sterilization provide greater assurance than the sterility test. This method of assuring sterility is termed parametric release and is defined as the release of a sterile product based on process compliance

with physical specifications. Parametric (real time) release is acceptable to regulatory authorities, if justified, for fully validated terminal sterilization processes recommended by the *European Pharmacopoeia*.

Validation of a sterilization process

The *British Pharmacopoeia* (British Pharmacopoeia Commission, 2020a) states:

The achievement of sterility for any one item in a population of items submitted to a sterilisation process can neither be guaranteed nor demonstrated. It is essential to study the effect of the chosen sterilisation procedure on the product (including its final container) to ensure its effectiveness and the integrity of the product, and to validate the procedure before it is applied in practice.

Clearly this statement points out that testing for sterility is not enough and a suitable production process should be appropriately validated. Any changes in the sterilization procedure (i.e. change in sterilization process, product packaging or load) require revalidation. For pharmaceutical preparations, Good Manufacturing Practices (GMP) have to be observed for the entire manufacturing process, not just the sterilization procedure.

The process of validation requires that the appropriate documentation is obtained to show that a process is consistently complying with predetermined specifications. International organizations such as the International Organization for Standardization (<http://www.iso.org>) and the Food and Drug Administration in the USA (<http://www.fda.gov>) provide detailed documentation for the validation of sterilization of health care products or medical devices with various processes (e.g. steam, radiation and gaseous sterilization). For the validation of sterilization processes, two types of data are required: commissioning data and performance qualification data (Box 17.3). Commissioning data refer mainly to the installation and characteristics of the equipment, and the performance data ensure that the equipment will produce the required sterility assurance level. The performance qualification data can be divided into physical and biological performance data (Box 17.3).

Obtaining biological performance data is required for the validation and revalidation of the sterilization process

Box 17.3 Information required for the validation of a sterilization process**Commissioning data**

- Evidence that the equipment has been installed in accordance with specifications
- Equipment is safe to use
- Equipment functions within predetermined limits.

Performance qualification data

- Evidence that equipment will produce a product with an acceptable assurance of sterility
- Physical performance qualification — evidence that the specified sterilization conditions have been met throughout the sterilization cycle:
 - the tests performed depend on the sterilization process
 - data should be generated from the worst region in the sterilizer
 - the data generated should also show no detrimental effect on the product and its packaging
- Biological performance qualification — evidence that the specified sterilizing conditions deliver the required microbiological lethality to the preparation/product:
 - makes use of biological indicators
 - data are not required if the process is well defined (e.g. use of *F* value).

for new preparations, new loads and new sterilization regimens and is usually not used routinely except when the sterilization conditions are not well defined (e.g. gaseous sterilization) or with nonstandard methods. The use of biological indicators (discussed in the next section) requires a good knowledge of the inactivation kinetics (e.g. *D* value) for a given process. Performance qualification data must be re-evaluated following a change to the preparation or product and its packaging, the loading pattern or the sterilization cycle.

Process indicators

For all methods of sterilization, it is essential that the equipment used works correctly. Routine tests are carried out to demonstrate that all parts of the sterilizer have been correctly installed (installation qualification) and that they operate properly, with sterilizing conditions reaching every part of the load (operation qualification; McDonnell, 2007). The test methods used vary according to the sterilization method and may involve the use of physical indicators, chemical indicators and biological indicators.

Physical indicators measure parameters such as heat distribution (i.e. temperature) by thermocouples, pressure variation by gauges or transducers, gas concentration, steam purity, relative humidity by hygrometers or direct calorimetry, delivered dose and time exposure. Sensors must be maintained and calibrated regularly. They are usually the first indicators of a problem with a sterilization process. Sensors maintenance and calibration are essential to ensure the validity of parametric release (Berube et al., 2001).

Chemical indicators vary depending on the sterilization method but essentially they all change in physical or chemical nature in response to one or more parameters. There are several types of chemical indicators (Fig. 17.5); temperature-specific indicators just show whether a specific temperature has been reached (single-variable indicators), whereas multi-parameter/multivariable indicators can measure more than one variable at a time, e.g. heat and time or gas concentration and time, or time, steam and temperature.

Process indicators demonstrate that an indicator has gone through a process but they do not guarantee that sterilization was satisfactory. A common example is autoclave tape (single end-point indicator), which reflects the conditions inside the chamber environment but is not able to demonstrate that an item has been sterilized. More specific indicators, such as the 'Bowie–Dick test', are used to monitor air removal from autoclaves. They must be used in the first cycle of the day as an equipment function test (McDonnell, 2007). The standardized test pack is placed in the centre of porous load sterilizers, and if the process is correct (i.e. air removal is appropriate), uniform colour change occurs across the test package (Fig. 17.6).

Common chemical indicators are quantitative and indicate a combination of critical variables within a process. This is the case with dosimeters (e.g. Perspex), which gradually change colour on exposure to radiation. The performance of chemical indicators can be altered by the storage conditions before and after use and by the method used.

Biological indicators consist of a carrier or package containing a standardized preparation of defined microorganisms of known resistance to a specific mode of sterilization (Berube et al., 2001; see Fig. 17.5). The carriers used are usually made of filter paper, a glass slide, stainless steel or a plastic tube. Some versions incorporate ampoules containing a growth medium to minimize false positives. The carrier is covered to prevent deterioration or contamination while still allowing entry of the sterilizing agent (British Pharmacopoeia Commission, 2020a). Different organisms are used for different processes (Table 17.6), but biological indicators usually consist of bacterial spores. In excess of between 10^5 and 10^7 spores are used, the



Fig. 17.5 Examples of chemical and biological indicators. **(a)** Multiparameter (time, steam and temperature) indicators. **(b)** Sterilization control tubes. **(c)** *Geobacillus stearothermophilus* biological indicators.

recommended number being dependent on the sterilization method being assessed. After exposure to the sterilization process, the indicators are removed aseptically and incubated in suitable media to detect the presence of surviving microorganisms. If no growth occurs, the sterilization process is said to have had sufficient lethality (Berube et al., 2001).

Testing filtration efficacy

Compared with other sterilization methods, the potential risk of failure is higher for filtration sterilization. This means that it may be advisable to add an extra prefiltration stage using a bacteria-retentive filter. Confidence in the filters used is of prime importance during filtration sterilization. Each batch of filters is tested to ensure that they meet the specifications for release of particulate materials, mechanical strength, chemical characteristics

(e.g. oxidizable materials and leaching of materials) and filtration performance. The methods for testing filtration performance involve either a challenge test (which is destructive so cannot be conducted on every filter in a batch) or an integrity test (Walsh & Denyer, 2013).

The microbial challenge test is used to demonstrate that a filter is capable of retaining microorganisms. This is normally performed with a suspension of at least 10^7 colony-forming units (see Chapter 14) of *Brevundimonas diminuta* per square centimetre of active filter surface. *Brevundimonas diminuta*, is a small ($0.2\ \mu\text{m}$ to $0.9\ \mu\text{m}$) Gram-negative short rod that is a natural choice for this test because of its size and because it was originally isolated from contaminated filtered solutions (Levy, 2001). After filtration of a bacterial suspension prepared in tryptone soya broth, the filtrate is collected and incubated at $32\ ^\circ\text{C}$. For filtration with a $0.1\ \mu\text{m}$ nominal pore size, *Acholeplasma laidlawii* may be used.

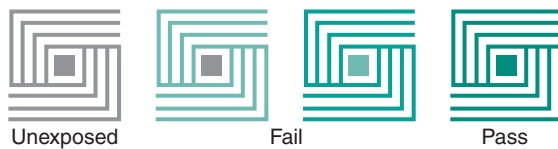


Fig. 17.6 Bowie–Dick test pack used to monitor air removal from steam sterilizers; a uniform colour change indicates sufficient steam penetration.

Table 17.6 Organisms used as biological indicators for sterilization

Sterilization process	Spores used as a biological indicator
Dry heat	<i>Bacillus atrophaeus</i> ATCC 9372. NCIMB 8058 or CIP 77.18
Moist heat	<i>Geobacillus stearothermophilus</i> ATCC 7953. NCTC 10007. NCIMB 8157 or CIP 52.81
Ethylene oxide	<i>Bacillus atrophaeus</i> ATCC 9372. NCIMB 8058 or CIP 77.18
Radiation	<i>Bacillus pumilus</i> ATCC 27142. NCTC 10327. NCIMB 10692 or CIP 77.25
Filtration	<i>Brevundimonas diminuta</i> ATCC 19146, NCIMB 11091 or CIP 103020. <i>Acholeplasma laidlawii</i> ATCC 23206

Integrity tests are used to verify the integrity of an assembled sterilizing filter before use and to confirm integrity after use. The tests used must be appropriate to the filter type and the stage of testing and may include bubble point tests, diffusive flow measurement or water-intrusion testing. The bubble point test is the oldest and one of the most widely used nondestructive tests. It measures the pressure (bubble-point pressure) needed to pass gas through the largest pore of a wetted filter. In practice, the pressure required to produce a steady stream of gas bubbles through a wetted filter is often used as the bubble point. The basis of the test relies on the pores through the filter resembling uniform capillaries passing from one side to another. If these capillaries become wet, then they will retain liquid via surface tension, and the force needed to expel the liquid with a gas is proportional to the diameter of the capillaries (pore diameter). The main limitations of this technique are that it is reliant on operator judgement

and on the holes in the filter being perfect uniform capillaries (Walsh & Denyer, 2013).

Diffusion rate tests are especially useful for large-area filters. They measure the rate of flow of a gas as it diffuses through the water in a wetted filter. The pressure required to cause migration of the gas through the liquid in the pores can be compared with data specified by the filter's manufacturer to establish if the filter has defects (Levy, 2001).

Monitoring decontamination

The possibility of transmission of Creutzfeldt–Jakob disease (CJD) highlighted the importance of protein removal from previously contaminated high-risk instruments. Visual inspection and ninhydrin or modified *o*-phthalaldehyde (OPA) methods may not be as sensitive as newer methods. Scanning electron microscopy (SEM) and energy-dispersive X-ray spectroscopy (EDX) analysis are not practical for health care professional use, so recent methods have concentrated on using fluorescent reagents coupled with digital imaging; for example, epifluorescence differential interference contrast microscopy (EFDIC) and epifluorescence scanning (EFSCAN) (Baxter et al., 2014). Commercial products such as ProReveal can produce a high level of confidence in the effectiveness of washer disinfectant processes. More information can be found in ISO 15883-1.

Limitations of sterilization methods

Sterilization processes can involve some extreme conditions, such as high temperatures, high pressure, a vacuum and pressure pulsing, or the use of toxic substances, which can damage the product and/or its packaging. The alteration of a pharmaceutical preparation might lead to reduced therapeutic efficacy or patient acceptability, and damage to the container might lead to the poststerilization contamination of the product. There needs to be a balance between acceptable sterility assurance and acceptable damage to the product and container. Knowledge of the preparation and packaging design and the choice and understanding of the sterilization technologies help in making the appropriate selection to achieve maximum microbial kill while decreasing the risk of product and packaging deterioration.

Nevertheless, each sterilization technology has its limitations (Table 17.7). Limitations associated with established and recommended procedures are usually linked to the nature of the process (e.g. heat, irradiation), whereas newer technologies tend to suffer from a lack of reproducibility.

Table 17.7 Limitations of sterilization processes

Sterilization processes	Limitations
Heat sterilization	
Steam	Heat; damage to preparation Vapour; damage to the container (wetting of final product, risk of contamination after sterilization) Pressure; air ballasting; damage to the container
Dry heat	Heat; damage to preparation Potentially longer exposure time needed
Gaseous sterilization	
Ethylene oxide	High toxicity: risk to the operator Decontamination required after the process Explosive: risk to the operator Slow process ^a Many factors to control
Formaldehyde	High toxicity: risk to the operator Damage to some materials (e.g. materials made from cellulose) Decontamination required after the process Slow process ^a Many factors to control
Radiation sterilization	
γ -radiation	Risk to the operator Water radiolysis: damage to the product Discolouration of some glasses and plastics (including PVC), destructive process may continue after sterilization has finished Liberation of gases (e.g. hydrogen chloride from PVC) Hardness and brittleness properties of metals may change Butyl and chlorinated rubber are degraded Changes in potency can occur High costs
Particle radiation	β -radiation: risk to the operator Water radiolysis: damage to the product Poor penetration of electrons exacerbated by density of product Significant product heating may occur at high doses High costs
Chemosterilants	
Peracetic acid	Corrosiveness: damage to the product/device Activity: reports of microbial resistance
Filtration sterilization	
	Not efficient for small particles (viruses, prions) Requires strict aseptic techniques Integrity of membrane filter Growth of microbial contaminants in depth filter Shedding of materials from depth filter
^a Relative to moist heat sterilization. PVC, polyvinyl chloride.	

Summary

The achievement of sterility is a complex process that requires proper documentation. Sterility in the microbiological sense cannot be guaranteed. Therefore the sterility of a product has to be assured by the application of an appropriate validation process. It is important that the sterilization methodology is compatible with the preparation or product, including its final container or packaging, and combines effectiveness and the absence of detrimental effects. Although not described in detail in this chapter, the choice of the container/packaging must allow the optimum sterilization to be applied and assure that sterility is maintained after the process. Sterilization occurs at the end of manufacturing but it does not replace or permit a relaxation of the principles of GMP. In particular, the microbiological quality of ingredients for pharmaceutical preparations and the removal/reduction of bioburden must be monitored. Monitoring the critical

parameters of the sterilization process will ensure that the predetermined conditions (during validation) are met. The lack of validation, or failure to follow a validated process, carries the risk of a nonsterile product, deterioration and possible infection.

Where possible, terminal sterilization is the method of choice. Processes that are fully validated allow the possibility of parametric release of the preparation/product and hence their rapid commercialization, since sterility testing, and the delay it incurs, might not be necessary.

A clear understanding of the method, the product to be sterilized (including its packaging), the validation process and the overall documentation required is therefore necessary to carry out a successful sterilization.

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